

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-02

AI Multi-Modal Clinical Diagnostics & Triage Accelerator

Domain: Healthcare AI / Computer Vision / Edge Inference

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Develop an edge-capable AI inference suite that integrates clinical imaging (CT/X-ray), lab biomarkers, and EHR telemetry for instantaneous high-acuity triage and automated differential diagnosis.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] Multi-modal neural network fusing DICOM imaging with unstructured text clinical notes.

[REQ-2] Local ONNX / TensorRT edge model deployment for offline clinic resilience.

[REQ-3] Explainable AI heatmap overlays (Grad-CAM) justifying clinical alerts to physicians.

[REQ-4] HIPAA and Indian Digital Personal Data Protection (DPDP) compliant data anonymization.

[REQ-5] High-throughput HL7 / FHIR protocol ingestion interface.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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